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A remote sensing evidence on the marginality, stagementation and spatiotemporal heterogeneity of vegetation evolution characteristics in the Yinshan Mountains, China: Based on PKU GIMMS NDVI (1984–2022)

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ABSTRACT

This study focuses on a typical Ecological Vulnerable Area (EVA)-Yinshan mountains (YSMs), using PKU GIMMS NDVI as a vegetation growth indicator, to explore the spatiotemporal heterogeneity and driving mechanisms of vegetation in in desertified steppe ecosystems (DSEs) of arid and semi-arid regions. The Theil-Sen (T-S) Median trend analysis & Mann-Kendall (M-K) trend test, Hurst index, correlation analysis, partial correlation analysis, residual analysis and bivariate spatial autocorrelation (Bi-SA) analysis were used for quantitative analysis. Findings revealed: (1) Vegetation distribution had spatial heterogeneity, with a stepwise distribution from southeast to northwest; (2) Vegetation growth had temporal stagementation, with "stable-plunge-rise" pattern; (3) Climate trend displayed "warmer and wetter", with growth rates of 0.045 °C/a and 0.558 mm/a, respectively; (4) Vegetation response to precipitation was sensitive with strong spatial correlation (*Moran's* $I = 0.88$, $P < 0.01$); (5) Influence of human activities on the growth of vegetation was weak. The study emphasizes that the overall ecological quality of the YSMs has long been low. In recent decades, despite some improvements, the ecosystem remains extremely sensitive, fragile and volatile. Consequently, there is an urgent need to strengthen eco-environmental protection and management.

1. Introduction

Desertified steppe ecosystem (DSE) is a typical native ecosystem in arid and semi-arid regions (ASRs) of China, where the eco-environment is the most fragile and the highest concentration of poverty, and it also serves as a source and major occurrence area for sandstorms (Cheng et al., 2020). In recent decades, due to the frequency of climatic extremes and the intensification of human activities, the balance of ecosystems has been continuously disrupted (IPCC and Geneva, 2023), especially in ecological vulnerable area (EVA), as a result of which regional ecosystem issues have increasingly become focal point in the fields of ecology and global change science (Jiang et al., 2022; Liu et al., 2021). The United Nations has also incorporated combating climate change (SDG13) and desertification (SDG15) into its 17 Sustainable Development Goals for the 2030 Agenda (Pekmezovic et al., 2019).

Vegetation, as a comprehensive "indicator" for evaluating ecological changes and a crucial "linkage" between the land surface and the atmosphere (Ren et al., 2023; Su et al., 2022), is often considered as a sensitive index of climate change and used to comprehensively reflect the characteristics and quality of terrestrial ecosystems (Jiang et al., 2024; Tuoku et al., 2024). Remote sensing data have become an vital data source for long-term monitoring ecological changes because of excellent spatiotemporally continuity (Li et al., 2021a,b; Xiu et al., 2021). Normalized Difference Vegetation Index (NDVI) enhances the contrast between vegetated and non-vegetated areas in a non-linear stretching manner, which can effectively reflect the health status of vegetation, and is one of the most widely applied evaluation indexes in remote sensing vegetation monitoring to date (Li et al., 2021a,b; Liu et al., 2023). The long-term NDVI time series products offered by remote sensing technology, such as GIMMS (Global Inventory Modelling and

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Mapping Studies) NDVI3g (1981–2012) (Pinzon and Tucker, 2014), MODIS NDVI (2000–present) (Seguini et al., 2024) and SPOT NDVI (1999–present) (Bai et al., 2019), provide a reliable data foundation to reveal vegetation dynamics over extended time periods. Among them, the newly released PKU GIMMS NDVI (1982–2022) dataset (Li et al., 2023) from Peking University has a comparable spatial resolution with GIMMS NDVI3g (8 km × 8 km), which has a slightly lower spatial resolution but the longest time series compared to MODIS and SPOT NDVI datasets. It is more suitable for ecological studies of large- & meso-scale geographic units, adequately reflects the dynamic of vegetation over multi- life cycles.

The Yinshan mountains (YSMs) are located on the southern edge of the Mongolian Plateau and north of the Hetao Plain in China. As the primary body of wind-breaking and sand-fixing area in northern China, they play a crucial role in blocking wind and retaining water for the Beijing-Tianjin region. Furthermore, this area represents a classic arid & semi-arid climate transition zone (Yang et al., 2023b), an desert & steppe ecological transition zone (You et al., 2021) and an agro-pastoral ecotone (Liu et al., 2024b), with a monsoon climate zone on the southeast side and deep inland non-monsoon climate zone (*i.e.*, continental climate zone) on the northwest side, occupying an extremely important ecological position in arid-semi-arid ecosystems in northern China (Bai et al., 2020). Predicting the future evolution trend of regional climate and ecosystems, effectively assessing the risks faced by eco-environment construction, and comprehensively carrying out research on climate change and ecosystem response in sensitive areas are of great strategic significance in terms of location and ecology.

To date, numerous scholars have conducted research on the ecological status of the YSMs. Xia et al. (2007a) and Zhao et al. (2017b) investigated vegetation evolution in the north foot of Yinshan mountains (NF-YSMs) during 1981–2003 and 2000–2010, respectively. Fan et al., (2007), using the NASA/GIMMS NDVI dataset, analyzed the response of vegetation patterns to climate change in the NF-YSMs from 1982 to 2003, finding a strong correlation between vegetation growth and precipitation. Zhang et al. (2024a,b) evaluated the ecological quality of the Tabu river and Aibugai river basins in the NF-YSMs from 1986 to 2022 utilizing a modified ecological index. Additionally, Feng, (2021) and Chen, (2016) studied plant diversity and distribution patterns of the YSMs, demonstrating that the region is an integrated ecosystem with rich biodiversity. However, previous studies still present the following issues: (1) The study period is relatively short, which is not conducive to a longer time-series assessment of ecosystem sensitivity; (2) Most of the research only focuses on the NF-YSMs, lacking spatial linkage between vegetation growth and climate factors on both northern and southern sides of the YSMs; and (3) Existing studies often consider the climate factors as the principal factor for vegetation dynamic changes, ignoring the co-driving role of natural factors and human activities.

In view of this, taking the vegetation evolution characteristics of the YSMs as an entry point, based on the PKU GIMMS NDVI dataset from 1984 to 2022, along with the meteorological and the land cover data of the same period, with the help of Theil-Sen (T-S) Median trend analysis & Mann-Kendall (M-K) trend test, Hurst index, correlation analysis, partial correlation analysis, residual analysis, and bivariate spatial autocorrelation (Bi-SA), this paper deeply analyzed the vegetation evolution pattern and sustainability characteristics at different spatial and temporal scales in the YSMs, explored the spatial correlation model between NDVI and climate factors, and quantified the contribution of main driving factors such as climate factors and human activities. The innovative contributions of the paper are: (1) The long-period spatio-temporal consistency global dataset used in this paper can fully characterize the spatiotemporal changes of land surface vegetation at large- & meso-scale; (2) The multi-method coupled analysis based on general vegetation monitoring indicators enhances the universality, facilitating widespread promotion and peer comparison; and (3) Focusing on a typical ASRs in China provides a reference for assessing ecosystem quality in ASRs nationwide. The main purpose is to inform decision-making basis for ecological restoration and biodiversity conservation in ASRs, while providing data support for the achievement of SDG13 (*i.e.*, Take urgent action to combat climate change and its impacts) and SDG15 (*i.e.*, Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss) in the United Nations 2030 Sustainable Development Goals (SDGs) (Gong, 2019).

2. Materials and methods

2.1. Overview of the study area

The YSMs (40° ∼ 43° N, 104° ∼ 117° E), located in northern China and central Inner Mongolia Autonomous Regions with an overall east–west orientation, belong to an ancient fault block mountain system (Lin et al., 2013). From west to east, it consists of Langshan mountain, Wulashan mountain, Daqingshan mountain, Huitengliang mountain and Damaqun mountain respectively, stretching 1,200 km in length and 50 ∼ 100 km in width. The highest peak, Mt. Huhebashige in the Langshan mountain, reaches an elevation of 2,364 m (Fan et al., 2024). The YSMs hold a unique ecological and geographical location as a classic agro-pastoral ecotone in China, with farming and nomadic civilizations to the south and north of the mountains separately (Wu, 2023). At the same time, it is the watershed between the non-monsoon and monsoon zones in China, subjected to the co-influence of temperate continental and monsoon climate (Chen and Shen, 2021). Furthermore, the region is highly susceptible to human interference and natural disasters, making the ecosystem extremely fragile; in particular, the large wind-eroded sandy lands at the NF-YSMs are the major source of sandstorms in China (Liu et al., 2024a), which poses a serious threat to the ecological security of northern China. Therefore, it is imperative to carry out research on long-term eco-environmental dynamic monitoring of the YSMs.

The northern and southern sides of the YSMs are extremely asymmetric in geomorphological form. The northern terrain transitions gently to the Mongolian Plateau, while the southern edge is steep fault cliffs, plummeting to the Hetao Plain with a drop of over a thousand meters (Yuchi et al., 2021; Zhang et al., 2020). This constitutes a mountainous ecological intertwining belt around the southern edge of the ASRs (Yang et al., 2023a), becoming a natural ecological barrier in northern China. The research used ArcGIS® 10.1 to map the closed vector boundaries of the YSMs based on the 1,200 m south and 1,400 m north contours (Fig. 1).

2.2. Data sources and preprocessing

2.2.1. NDVI dataset

NDVI dataset was obtained from PKU GIMMS NDVI v1.2, a long-term NDVI time series product with a spatial resolution of 8 km released by Peking University. This dataset was a half-monthly vegetation index synthesized employing the Maximum Value Composition (MVC) method, offering global NDVI data for 1982–2022 (Li et al., 2023). Since the half-monthly NDVI composited by MVC have basically eliminate the atmospheric effects on the NDVI magnitude, it was allowed to get monthly & annual NDVI datasets directly based on the upper and lower two semi-monthly data on this basis, including scale factor adjustments.

2.2.2. Meteorological dataset

Temperature and precipitation datasets were sourced from the National Tibetan Plateau/Third Pole Environment Data Center, providing "1-km monthly mean temperature dataset for China (1901–2022)" (Peng, 2019) and "1-km monthly precipitation dataset for China (1901–2022)" (Peng, 2020). Their spatial resolution was 0.0083333° (approximately 1 km), with units of 0.1 °C and 0.1 mm respectively.

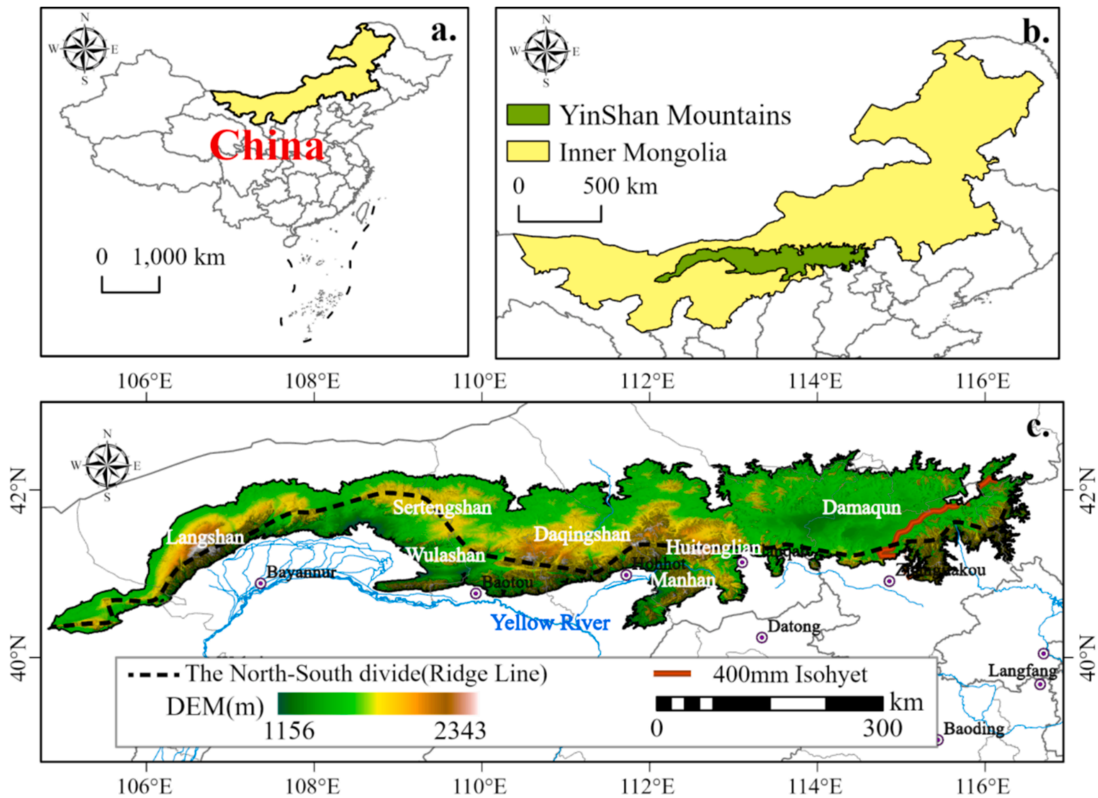

Fig. 1. Geographic location and topography of the study area.

2.2.3. Nighttime-light dataset

PANDA (1984–2020) was released by the National Tibetan Plateau/Third Pole Environment Data Center (Zhang et al., 2021). It was computed based on the Nighttime Lights Convolutional Long Short-Term Memory (NTLSTM) network, and was the world's first artificial nighttime lights dataset for China from 1984 to 2020. In addition, this dataset had high data quality, and the model evaluation with the original images resulted in an *RMSE* of 0.73, a coefficient of determination (R^2) of 0.95, and pixel slope of 0.99. In this study, 2000–2000 were selected, and the digital number (DN) was 0 to 6300, which was a dimensionless quantity; the spatial resolution was 1 km.

2.2.4. Population dataset

Landscan Global dataset was obtained from Oak Ridge National Laboratory (ORNL), with a spatial resolution of 1 km and a time span of 2000–2023 (Sims et al., 2021). And 2000–2020 were selected for analysis in this study.

2.2.5. Grazing intensity dataset

The grazing intensity data were obtained from the Global Resources Data Cloud, a dataset covering the long-term high-resolution grazing intensity data of grasslands in seven pastoral area in western China (Inner Mongolia, Xinjiang, Tibet, Qinghai, Gansu, Ningxia and Sichuan Provinces) for the years 1980–2022 (Wang, 2024). The spatial resolution of the data was 0.1° for 1980–2000 and 1 km for 2001–2022.

2.2.6. Land cover data

Land cover data was annual land cover dataset (China Land Cover Dataset, CLCD) issued by Wuhan University with spatial resolution of 30 m × 30 m, containing land use information from 1985, 1990–2022 (Yang and Huang, 2021). In this study, 4 periods of land cover data, 1990, 2000, 2010 and 2020, were selected to extract impervious surfaces to exclude the built-up areas in the study area, so as to improve the scientific validity of the subsequent analyses.

2.2.7. Digital elevation model data

DEM data was obtained from the SRTM DEM (90 m × 90 m) dataset released by Geospatial Data Cloud. The 4 remote sensing images, srtm_57_04, srtm_58_04, srtm_59_04, and srtm_60_04, were selected and processed through mosaicking and clipping to acquire the DEM data for the study area.

In order to ensure the spatiotemporal consistency among the multi-source datasets (Table 1), this research batch preprocesses each type of data based on ArcGIS® 10.1. The following steps are included: Firstly, the basic mosaicking and the unified spatial reference system (GCS_WGS_1984) were carried out to ensure that the images can cover the whole study area well and have a unified spatial reference; secondly, the data from multiple sources were resampled, and all the raster data in

Table 1
Statistical tables of data sources.

No.	Type	Time Span	Spatial/ Temporal Resolution	Data Sources
1	PKU GIMMS NDVI	1982–2022	1/12°×1/ 12°/15d	https://zenodo.org
2	Temperature&Precipitation	1901–2022	1 km × 1 km/ Monthly	http://data.tpdc.ac.cn
3	Nighttime-light dataset	1984–2020	1 km × 1 km/Yearly	
4	Population dataset	2000–2023	1 km × 1 km/Yearly	https://landscan.ornl.gov
5	Grazing intensity dataset	1980–2022	1 km × 1 km/Yearly	https://gis5g.com
6	China Land Cover Dataset	1985, 1990–2022	30 m × 30 m/Yearly	https://zenodo.org
7	SRTM DEM	/	90 m × 90 m	https://www.Gscloud.cn

this study were interpolated to 8 km using the "Nearest Neighbor Interpolation Method" to match the spatial resolution of the NDVI products; finally, the definition of borders and clipping are carried out to resolve the misalignment of pixels and ensure the consistency of data boundaries.

2.3. Methods

This study aims to achieve three main objectives: (1) Spatiotemporal variation characteristics analysis. PKU GIMMS NDVI, meteorological, CLCD and DEM data were preprocessed, then, applied the T-S Median & M-K trend analysis method to assess changes in time-series NDVI data; (2) Sustainability characterization. Use the combined T-S Median trend analysis and Hurst index to predict NDVI change trends; (3) Driving factors analysis. Pearson's correlation coefficient, partial correlation coefficient, residual analysis and Bi-SA were used to investigate the interaction between meteorological factors, human factors and NDVI. A detailed technical flowchart of the study is presented in Fig. 2.

2.3.1. Theil-Sen Median trend analysis and Mann-Kendall trend test

The T-S Median trend analysis and M-K trend test are two non-parametric statistical test methods commonly used for vegetation NDVI trend analysis (Kumar et al., 2022; Sen, 1968).

The T-S Median trend analysis is less sensitive to outliers and more robust compared to simple linear regression, making it a more reliable non-parametric trend analysis method (Poppe Terán et al., 2023). Hence, it is particularly suitable for analyzing trends in large time-series datasets. The calculation formula is as follows:

$$NDVI_T = \mathrm{Median}\left(\left(NDVI_j - NDVI_i\right)/(j-i)\right), \forall j > i. \tag{1}$$

where n is the sequence length of the time series data, $1 < j < i < n$; *Median()* indicates the median of the slope, that is, the vegetation change trend. When $NDVI_T > 0$, it implies that the $NDVI$ shows an upward trend; conversely, it is in a downward trend.

As the T-S Median trend analysis can only detect the trend of change in the series but cannot test the significance of the results, it is commonly used in conjunction with the M-K trend test, which helps assess the significance of the change in the trend of the series. The formula for calculating the M-K test statistic Z on a pixel-by-pixel basis are as follows:

$$Z = \begin{cases} (S-1)\Big/\sqrt{Var(S)} & S > 0 \\ 0 & S = 0 \\ (S+1)\Big/\sqrt{Var(S)} & S < 0 \end{cases} \tag{2}$$

$$Var(S) = (n(n-1)(2n+5))/18 \tag{3}$$

$$S = \sum_{i=1}^{n-1}\sum_{j=i+1}^{n} sgn\left(NDVI_j - NDVI_i\right) \tag{4}$$

$$\mathrm{sgn}\left(NDVI_j - NDVI_i\right) = \begin{cases} 1 & NDVI_j - NDVI_i > 0 \\ 0 & NDVI_j - NDVI_i = 0 \\ -1 & NDVI_j - NDVI_i < 0 \end{cases} \tag{5}$$

where n is the number of the NDVI data in the time series, i, j represent the time series, and *sgn()* is the sign function. A two-sided trend test was used to obtain the critical value $Z_{1-\alpha/2}$ based on the normal distribution at a given significance level α. When $|Z| > Z_{1-\alpha/2}$, the trend is considered to vary significantly at the α level. Here, given a significance level of $\alpha = 0.05$, the critical value is $Z_{1-\alpha/2} = \pm 1.96$. when $|Z| > 1.65$, 1.96, and 2.58, it indicates that the trend passes the significance test with a confidence level of 90 %, 95 %, and 99 %, respectively.

2.3.2. Correlation analysis and partial correlation analysis

Correlation analysis can reflect the closeness of between two or more elements, and here, the Pearson correlation coefficient was utilized to evaluate the degree of response of NDVI to temperature and precipitation (Ye et al., 2017). The calculation formula is shown in Eq. (6):

$$R_{XY} = \left(\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})\right)\Big/\left(\sqrt{\sum_{i=1}^{n}(X_i - \overline{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i - \overline{Y})^2}\right) \tag{6}$$

where R_{XY} denotes the correlation coefficient between NDVI and temperature or precipitation; n is the length of the time-series data; i is the ordinal number of the time series; X_i and Y_i denote the NDVI and climate factors in the i^{th} year, $\overline{X}$ and $\overline{Y}$ denote the average of NDVI and climate factors, respectively.

Ecosystems are complex systems composed of multiple interacting

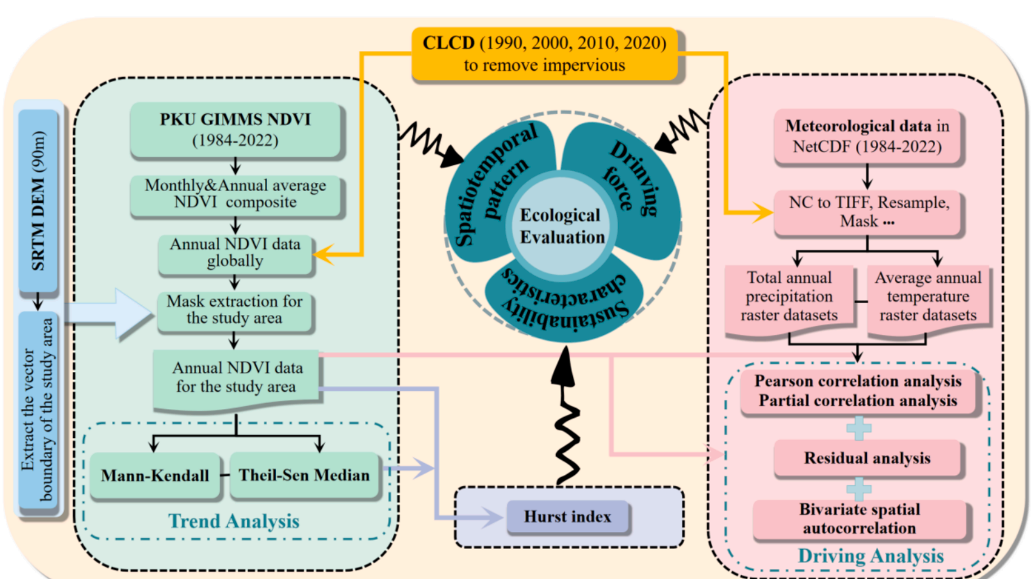

Fig. 2. Technical roadmap.

components, where changes in one element are often the result of interactions among several factors. However, the results of correlation analysis only involve the two variables of interest and can solely explain the overall association between them, offering a preliminary understanding of the degree of correlation. Based on this, this study further employed partial correlation analysis to control for another variable, thereby eliminating its disruptive impact and more effectively elucidating the net correlation between temperature or precipitation and NDVI (Ye et al., 2021). Additionally, partial correlation analysis allowed for further validation of whether these results remain significant when controlling for the influence of other variables, thereby enhancing the reliability and interpretative power of the findings (Yang et al., 2024). The partial correlation coefficient is defined in Eq. (7):

$$R_{XY,Z} = (R_{XY} - R_{XZ}R_{YZ}) \Big/ \left(\sqrt{(1 - R_{XZ}^2)(1 - R_{YZ}^2)} \right) \tag{7}$$

where X is vegetation NDVI, Y and Z are temperature and precipitation; $R_{XY,Z}$ represents the partial correlation coefficient between NDVI and temperature under the condition that precipitation is used as a control variable.

Both correlation coefficients are at [−1,1], with absolute values closer to 1 indicating a stronger correlation.

Here, significance testing was conducted using the *T-test*, with a significance level was set at $P = 0.05$.

2.3.3. Hurst index

The Hurst index is commonly used to predict future trends in long time-series data and is now widely used to quantitatively characterize the sustainability of long-term NDVI (Shrestha et al., 2024). The specific calculation process for Hurst index *(H)* includes the following several steps:

(1) Supposing there is a time series $\{NDVI_{(t)}\}$, $t = 1,2,3,\cdots,n$, the sequence of their averages $(\overline{NDVI})$ is defined as:

$$\overline{NDVI} = (1/\tau)\sum_{t=1}^{\tau} NDVI_{(t)} \qquad \tau = 1,2,\cdots,n \tag{8}$$

(2) Calculation of the cumulative deviation $(X_{(t,\tau)})$ using Eq. (9) as follows:

$$X_{(t,\tau)} = \sum_{t=1}^{\tau} \left(NDVI_{(t)} - \overline{NDVI}_{(t)} \right) \qquad \tau = 1,2,\cdots,n \tag{9}$$

(3) Calculation of the extreme deviation $(R_{(\tau)})$ and standard deviation $(S_{(\tau)})$ sequence using Eqs. (10) and (11) as follows:

$$R_{(\tau)} = \max_{1\leqslant t\leqslant\tau} X_{(t,\tau)} - \min_{1\leqslant t\leqslant\tau} X_{(t,\tau)} \qquad \tau = 1,2,\cdots,n \tag{10}$$

$$S_{(\tau)} = \left[(1/\tau)\sum_{t=1}^{\tau} \left(NDVI_{(t)} - \overline{NDVI}_{(t)} \right)^2 \right]^{1/2} \qquad \tau = 1,2,\cdots,n \tag{11}$$

(4) Finally, the Hurst index *(H)* is defined as:

$$R_{(\tau)}/S_{(\tau)} = (c\tau)^H \tag{12}$$

where c is the constant; τ is the time lag and the Hurst index *(H)* value is between 0 ~ 1. Specifically, when $0 < H < 0.5$, it indicates that the NDVI time series exhibits anti-sustainability, meaning that its future trend is opposite to its past; when $0.5 < H < 1$, it indicates that the NDVI time series has sustainable characteristics, meaning that its future trend aligns with its past; when $H = 0.5$, it represents random fluctuations and there is no long-term correlation.

2.3.4. Residual analysis

This study utilizes a multiple regression residual analysis method to quantitatively assess the respective contributions of climatic and non-climatic factors (primarily human activities) to changes in NDVI trends. The specific calculation steps are as follows: Firstly, a bivariate linear regression model is established with temperature and precipitation as independent variables and NDVI as the dependent variable, with the parameters being calculated. Secondly, based on temperature, precipitation and the model parameters, NDVI predictions are computed. Finally, the differences between observed and predicted NDVI values are calculated to obtain the NDVI residuals (Ibrahim et al., 2015; Jiang et al., 2017). Its formula is shown below Eqs. (13) and (14):

$$NDVI_{pre} = a \times T + b \times P + c \tag{13}$$

$$NDVI_{hum} = NDVI_{obs} - NDVI_{\mathrm{pre}} \tag{14}$$

where T and P represent temperature and precipitation; $NDVI_{pre}$ and $NDVI_{obs}$ represent predicted and observed NDVI values, respectively; a, b and c represent model parameters; and $NDVI_{hum}$ is residuals, which reflects the impact of human activities on the trend of NDVI. It is generally accepted that when $NDVI_{hum} > 0$, it indicates a promoting effect; conversely, it indicates an inhibiting effect.

2.3.5. Spatial autocorrelation analysis

Spatial autocorrelation is a statistical method used to measure the distributional characteristics and interrelationships of spatial data. Generally, there are two types: global and regional, which are employed to describe the overall spatial distribution pattern and the spatial heterogeneity of local regions, respectively (Chen, 2023).

Unlike the traditional univariate spatial autocorrelation, Bi-SA offers strong reliability and adaptability in describing the spatial relationship and dependency characteristics between two specific types of geographical elements (Tong et al., 2020). This study utilizes the Moran's I index *(I)*, obtained from bivariate global spatial autocorrelation, to evaluate the overall spatial correlation between NDVI and climatic factors in the YSMs. Particularly, $I \in [-1, 1]$, when $I > 0$, it indicates a positive spatial correlation; when $I < 0$, it indicates a negative spatial correlation and larger absolute values suggesting more pronounced clustering or dispersion trends; when I is close to 0, it indicates a random spatial distribution, *i.e.*, there is no correlation, which is calculated by:

$$I = \left(\sum_{i=1}^{n}\sum_{j=1}^{n} W_{ij}\left(x_i^p - \overline{x^p}\right)\left(x_j^q - \overline{x^q}\right) \right) \Big/ \left(S^2 \sum_{i=1}^{n}\sum_{j=1}^{n} W_{ij} \right) \tag{15}$$

Since global spatial autocorrelation can only judge whether clustering characteristics exist within the space as a whole, but lacks the function of reflecting the positional distribution characteristics of cluster types (Zhang et al., 2023), here, bivariate local spatial autocorrelation is used to describe the local spatial correlation pattern between NDVI and climate elements in the YSMs, with the following Eq. (16):

$$I_{Local} = \left(x_i^p - \overline{x^p}\right) \Big/ (S^p) \sum_{j=1}^{n} W_{ij}\left(\left(x_i^q - \overline{x^q}\right)/S^q\right) \tag{16}$$

where x_i^p and x_j^q are the attribute values; $\overline{x^p}$ and $\overline{x^q}$ is the average value; W_{ij} is the weight matrix established by spatial unit i, j $(i, j = 1,2,3,\cdots,n)$; S^p and S^q are the variances of the two types of attribute values.

3. Results

3.1. Characteristics of spatiotemporal variations of NDVI and climate factors

3.1.1. Characteristics of spatiotemporal variations at intra-annual scale

(1) Temporal variation characteristics. The intra-annual scale distribution of NDVI and climatic factors in the YSMs from 1984 to 2022 was researched on a monthly time scale. ① Both NDVI and climate factors in this region exhibited a distinct “single-peaked” features in the time dimension, with the peaks of temperature and precipitation occurring around July every year (21.04 °C, 69.67 mm), followed by a gradual decline, and then a minimum temperature of –14.75 °C in January of the following year (Fig. 3f); ② Precipitation was at a low level from December to February, with the lowest value being 3.91 mm. Apparently, there was a clear synchrony between rainfall and warmth, consistent with the climatic characteristics of the region (Fig. 3e); ③ Vegetation began to enter the greening period (growing season) in early June, its growth was most vigorous at the end of July reaching a $NDVI_{max}$ value of 0.36. By mid-September, vegetation entered the browning period (wilting season), and from December to February, nearly no vegetation cover is observed due to low-temperature stress (Fig. 3d). Overall, the temperature began to rise in late March, however, the vegetation did not show significant responsiveness to the temperature increase at this time. It was not until late May, when precipitation began to surge, that vegetation responded rapidly and positively, initiating the greening phase in early June.

On the inter-annual scale, the NDVI for the same periods each year fluctuates slightly, while the temperature shows an increasing trend. However, for precipitation, the precipitation in the same period shows irregular changes and fluctuates greatly. Notably, during the period from 2000 to 2010, there is a marked decrease in the high-value interval

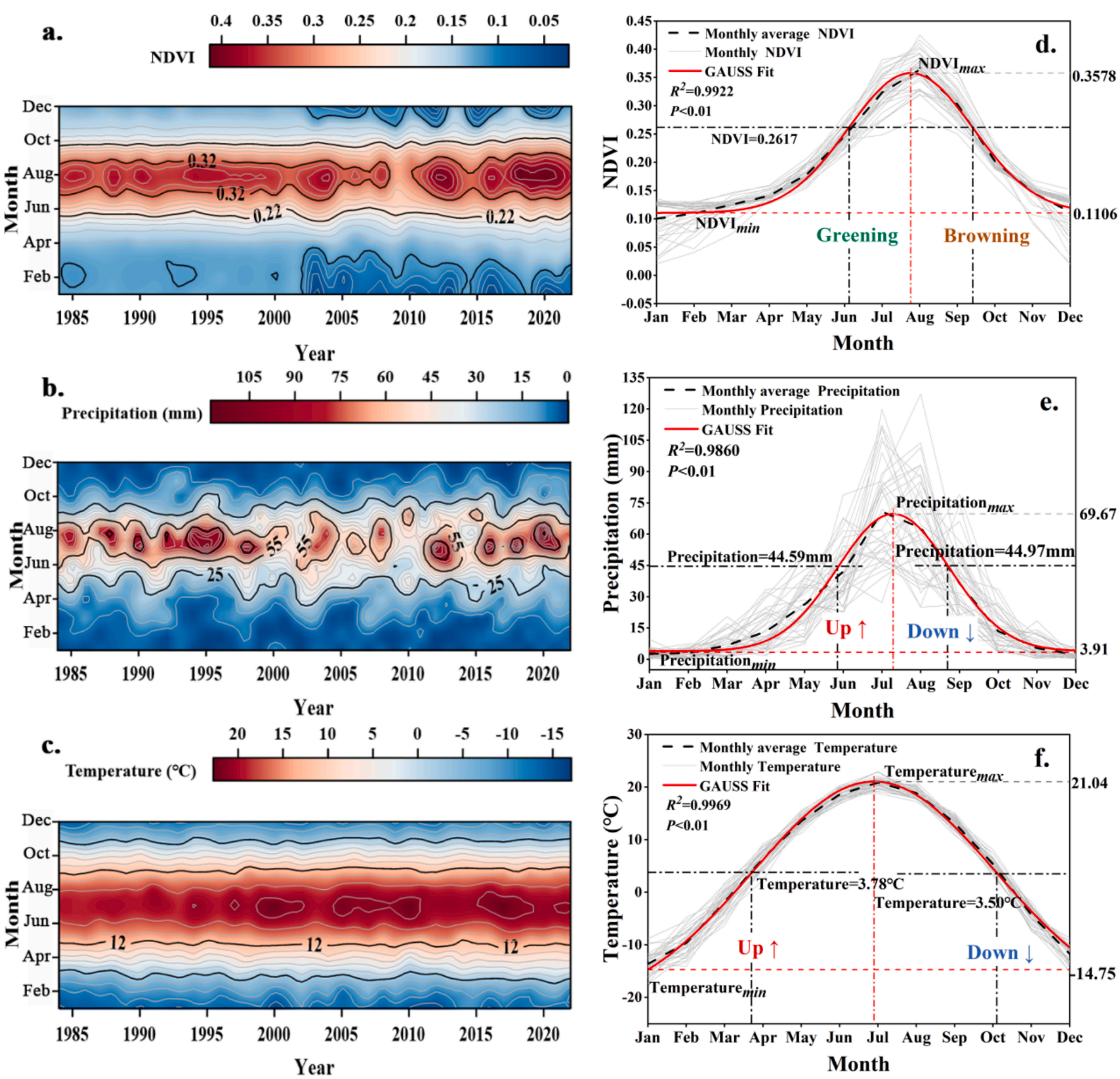

Fig. 3. Temporal change of NDVI and climate elements based on the monthly scale in the Yinshan mountains, 1984–2022: Halo map of monthly average NDVI, precipitation and temperature, **a–c**; Gaussian fit of monthly average NDVI, precipitation and air temperature for the same period of 39a, **d–f**. Note: Greening indicates that the vegetation is regreening (*i.e.*, Start Of growing Season, SOS); Browning indicates the browning of vegetation (*i.e.*, End Of growing Season, EOS). Up↑ indicates a rise in temperature or precipitation; Down↓ indicates a drop in temperature or precipitation.

compared to other years, characterized by a shortened rainy season and reduced maximum precipitation (Fig. 3a–c).

(2) Spatial variation characteristics. The intra-annual spatial distribution pattern of NDVI and climate factors in the YSMs from 1984 to 2022 is shown in Fig. 4. ① Since May, precipitation first reached the southeastern part of the Damaqun mountain under the influence of the Pacific southeast monsoon, triggering the greening of vegetation in this area; ② Precipitation in the YSMs was mostly concentrated between July and August. Along with the warming of the temperature, this period represented the optimum growth phase for vegetation. The greening range gradually spreaded to the areas of Daqingshan mountain, Huitengliang mountain, Wulashan mountain, reaching its largest extent by the end of August; ③ From September onward, vegetation entered the end of the growing season. With the successive harvesting of crops and the wilting of vegetation, the area of greening was once again narrowed down to the Damaqun mountain. ④ By the end of October, precipitation decreased sharply and temperatures dropped, marking the end of the vegetation growth season. The NDVI was at a low value as the region transitions into winter, during which there was almost no precipitation and vegetation cover.

In summary, from the perspective of spatial distribution, the patterns of vegetation and precipitation basically coincide with each other, both

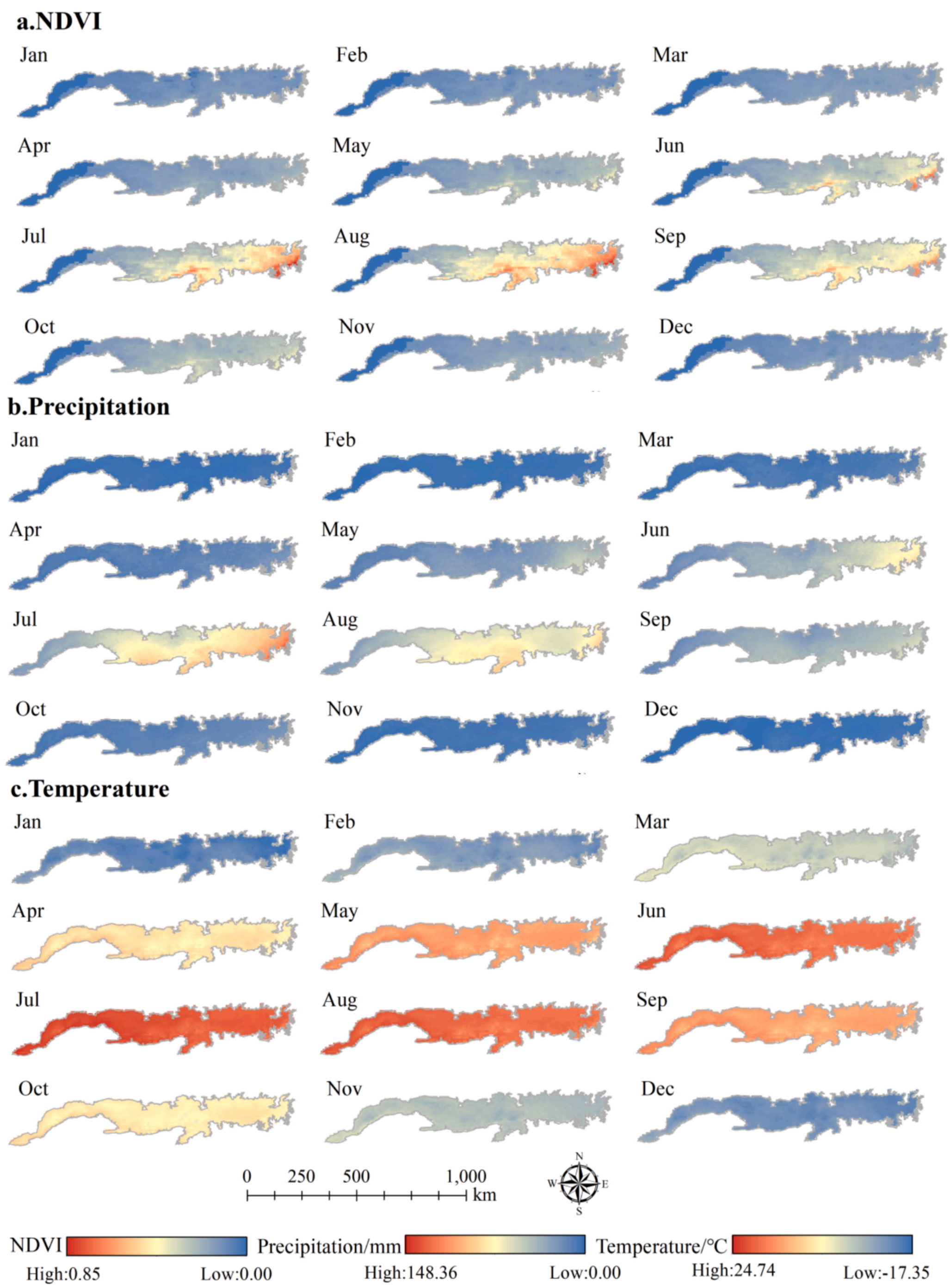

Fig. 4. Monthly vegetation and climate distribution patterns in Yinshan mountains from 1984 to 2022.

showing obvious regional differences; while the spatial differentiation of temperature is not obvious. In terms of temporal distribution, there is a synchronous relationship between temperature and precipitation, however, the vegetation growing season has a lagging relationship with the climate.

3.1.2. Characteristics of spatiotemporal variations at inter-annual scale

In this research, linear trend graphs were used to characterize the inter-annual variations of average annual NDVI, temperature and total precipitation in the YSMs from 1984 to 2022, as shown in Fig. 5.

On the inter-annual scale, the NDVI in the study area showed an overall slight decreasing trend ($Slope = -8.37 \times 10^{-5}/a$), but with obvious stage characteristics. Specifically, before 2003, the NDVI fluctuated between 0.18 ~ 0.21, with a smooth trend ($Slope = -2.51 \times 10^{-4}/a$); in 2005, it dropped sharply to a minimum value of 0.17, after which the NDVI exhibited a significant increasing trend ($Slope = 1.56 \times 10^{-3}/a$) and reached a peak of 0.22 in 2019.

As far as precipitation is concerned, the inter-annual precipitation throughout the study area showed a fluctuating upward trend ($Slope = 5.58 \times 10^{-1}\ mm/a$), with the maximum and minimum values occurring in 2003 (203.87 mm) and 2009 (375.79 mm), respectively. Notably, the precipitation was at a low level for three consecutive years from 1999 to 2001, measuring 250.81 mm, 248.44 mm, and 238.46 mm, respectively. China, as one of the countries with the highest frequency of droughts, is facing a major challenge as a result of aridification in the north, which is an important natural disaster in the country (He et al., 2016). Previous studies have shown that persistend drought in China, especially in the northern region, during 1999 to 2001 (Li et al., 2003; Zhao et al., 2017a), and the results of the research in this paper are highly compatible with it.

As far as temperature is concerned, there was a significant increase trend during the 39a ($Slope = 4.5 \times 10^{-2}$ °C/a), with an obvious warming effect. This phenomenon is consistent with the global climate change and is a positive response area to global warming.

The spatial distribution patterns of NDVI and climate factors in the YSMs (Fig. 6a–c) indicate significant spatial heterogeneity in NDVI, generally increasing from northwest to southeast. The most severe ecological level in the western Langshan mountain, which is basically align with the precipitation pattern of "more in the east than the west, more in the south than the north". The high temperature is mainly concentrated in the edge areas of the southwestern mountains with lower altitude due to the effect of the topographical influence. The special geographical position of the YSMs acts as a barrier to both the southerly cold flow and the northerly moisture; in addition, its northern slopes are affected by the Mongolia Plateau high-pressure airflow, resulting in low rainfall, year-round dry and the cold temperature. Conversely, the atmospheric precipitation at the southern slope is relatively sufficient, with rich in surface and groundwater, so, the climate is relatively warm and humid here. Therefore, this created a precipitation gradient that decreases from south to north and east to west, meanwhile, the southeastern part of the Damaqun mountain displayed the best vegetation growth status due to higher precipitation.

The T-S Median & M-K results were divided into 9 levels (Table 2) to quantitatively evaluate the trend characteristics of changes in the NDVI and climate factors in the YSMs from 1984 to 2022 (Fig. 6d–f).

Overall, NDVI in the YSMs exhibits, to vary degrees, a decreasing trend in high latitude areas, while rising gently in low latitude areas, and there is no change in the west of Langshan mountain, accounting for 53.69 %, 32.57 %, and 13.74 % of the total area, respectively. This indicats that vegetation degradation in the region remains serious, especially in the northern part of the YSMs, which is the most typical. Temperature showed a warming effect in the whole study area, with a 99.49 % share of extremely significant increases; precipitation only showed an non-significant downward trend only in some eastern areas, and the remaining areas were in an upward trend, accounting for 3.69 % and 96.31 %, respectively. Comprehensively, the areas with increasing trends in NDVI are primarily located south of the YSMs where water resources are relatively abundant, while the areas with decreasing trends in NDVI are mainly in the northern steppe zone where precipitation is scarce and the population is sparse.

3.1.3. Sustainability characteristics analysis

By coupling the Hurst index with T-S Median trend analysis, we further classified future NDVI change trends into 5 levels based on

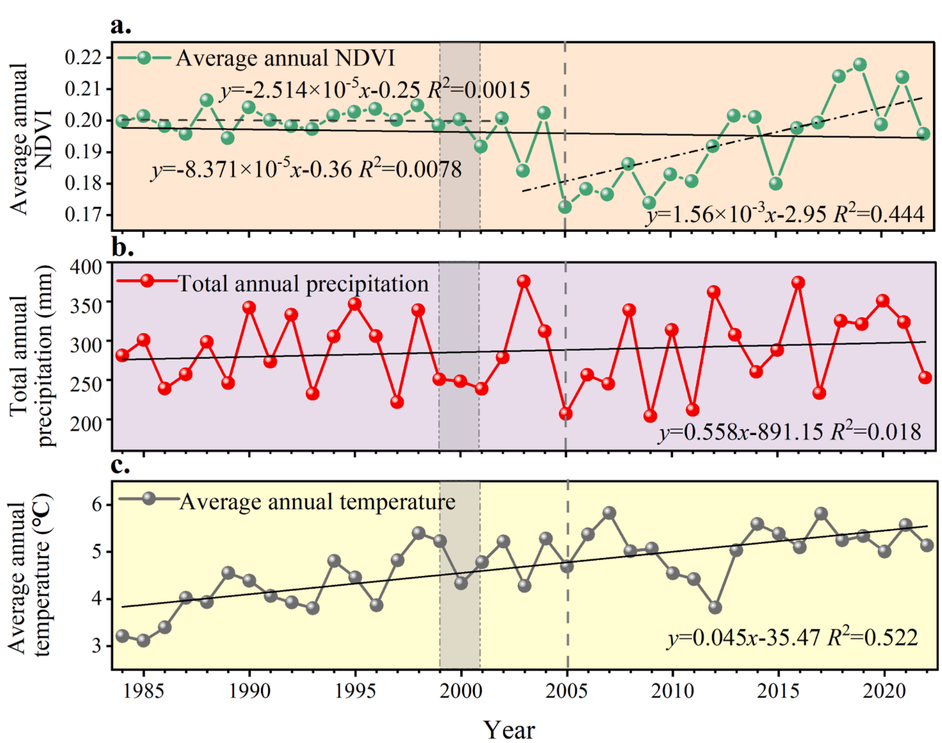

Fig. 5. Time series variation of elements in the study area, 1984–2022. **a.** NDVI; **b.** precipitation; **c.** temperature.

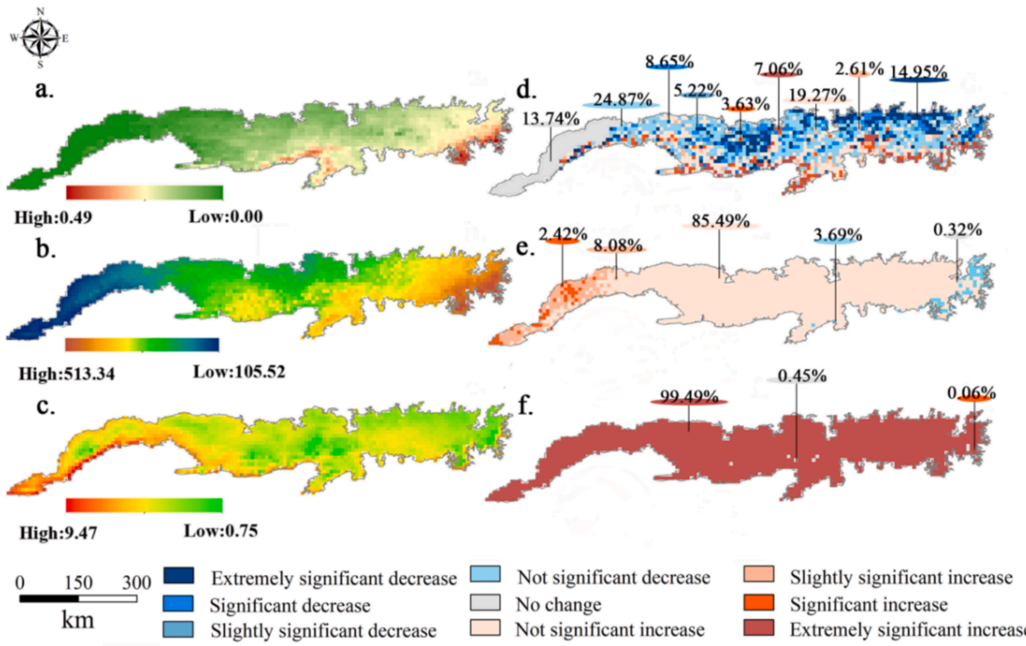

Fig. 6. Spatial distributions, **a–c.**, and trends in T-S&M-K, **d–f.**, of average annual NDVI and climate factors in the Yinshan mountains (1984–2022).

Table 2
Criteria for classification of levels for T-S & M-K.

$NDVI_T$	Z	Trends
$NDVI_T > 0$	$Z \le 1.65$	Not significant increase
	$1.65 < Z < 1.96$	Slightly significant increase
	$1.96 < Z < 2.58$	significant increase
	$Z < 2.58$	Extremely significant increase
$NDVI_T = 0$	Z	No change
$NDVI_T < 0$	$Z \le 1.65$	Not significant decrease
	$1.65 < Z < 1.96$	Slightly significant decrease
	$1.96 < Z < 2.58$	significant decrease
	$Z < 2.58$	Extremely significant decrease

Note: $NDVI_T$ indicates the trend of NDVI; Z indicates the statistical hypothesis testing.

Table 3
Criteria for levels of sustainability characteristics of NDVI based on the Hurst index.

$NDVI_T$	Hurst	Sustainability characteristics
$NDVI_T > 0$	$H > 0.5$	Sustainable increase
$NDVI_T < 0$		Sustainable decrease
$NDVI_T$	$H = 0.5$	Future trends are random
$NDVI_T > 0$	$H < 0.5$	Anti-sustainable increase
$NDVI_T < 0$		Anti-sustainable decrease

Note: $NDVI_T$ indicates the trend of NDVI; H indicates the future trend characteristics.

Table 3. This approach enables a deeper analysis of the sustainability characteristics of NDVI (Fig. 7).

Fig. 7a shows that the average Hurst index (H) is 0.47 and it ranged from 0.28 to 0.68 from 1984 to 2022. Specifically, $H > 0.5$ covered 28.05 %, $H < 0.5$ covered 54.71 % and the remaining 17.24 % was $H = 0.5$, indicating a strong anti-persistence in future vegetation NDVI in the YSMs.

Fig. 7b shows that the area with improving trends accounts for 35.56 %, mainly located in the northern of the YSMs, and the distribution is relatively concentrated; the degraded trends cover 20.61 %, primarily distributed in the southern of the Daqingshan mountain; the southern of Damaqun mountain and the southern edge of the YSMs, such as Wulashan mountain, Manhan mountain and Huitengliang mountain, forming a narrow pattern; sustainable decrease and sustainable increase account for 18.13 % and 12.02 %, respectively, and they are scattered sporadically in the Daqingshan mountain and Seertengshan mountain; random pattern makes up 13.68 %, predominantly in Langshan mountains, where the land cover is mostly barren and temperate steppes.

3.2. Response mechanism of NDVI to climate factors and human activities

3.2.1. Spatiotemporal relevance analysis

The spatial correlation between NDVI and climate factors (temperature & precipitation) in the YSMs was generally assessed based on the bivariate global Moran's I. Fig. 8a displays that the average Moran's I of NDVI and temperature is −0.46 ($P < 0.01$), showing negative spatial correlation, *i.e.*, NDVI and temperature are discretely distributed. Oppositely, Fig. 8b displays that the average Moran's I of NDVI and precipitation is 0.88 ($P < 0.01$), which is close to 1, with points concentrated in the first and fourth quadrants. This indicated a strong positive spatial correlation between NDVI and precipitation, with a clustered geographical distribution.

Further, based on the bivariate local spatial autocorrelation, the Local Indicator of Spatial Association (*LISA*) cluster map of spatial relationships between the binary variables were obtained (Fig. 9) and the long-term dynamic shifts in clustering patterns in four different periods were quantified using the transition matrix (Fig. 10). The clustering patterns were classified into five categories: "High-High (H-H)" representing areas with high climate factor values and high vegetation cover; "Low-Low (L-L)" representing areas with low climate factor values and low vegetation cover; "Low-High (L-H)" for low climate factor values with high vegetation cover; "High-Low (H-L)" for high climate factor values with low vegetation cover; and "Not Significant (H-L)" indicating that the spatial clustering feature is not apparent.

In terms of spatial distribution, the cluster patterns of NDVI with temperature and precipitation showed significant spatial differentiation. The four cluster patterns of H-H, L-L, H-L, and L-H were all distributed on the east and west sides of the YSMs with the boundary of Non-significant. The results of the research on precipitation and NDVI showed obvious gradient characteristics: H-H clusters were distributed in the western Damaqunshan mountain and along the southern edge of

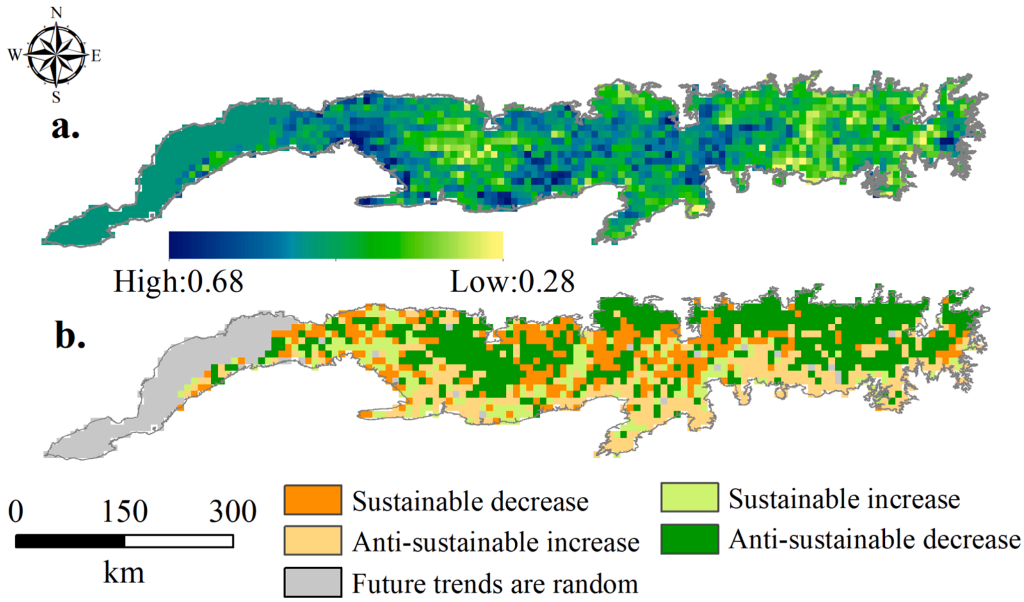

Fig. 7. Spatial distributions of the Hurst index of NDVI and future trend in the Yinshan mountains. **a.** Hurst index; **b.** Sustainability characteristics.

the Manhanshan mountain, where vegetation growth conditions were favorable, mostly temperate steppes and forest, with relatively lush vegetation, there was even a small piece of the L-H cluster located near Manhanshan mountain; the L-L clusters were widely found in the eastern Langshan mountain, and the vegetation was relatively sparse in this area. The temperature and NDVI, on the other hand, showed a slight patchy distribution, with the H-L clusters surrounded the L-L clusters in the Langshan mountain, while the L-H and H-H clusters were adjacent to each other, highly overlapping with the locations of the Damaqunshan mountain, the Manhanshan mountain, and Huitengliang mountain.

In terms of temporal characteristics, the results of precipitation and NDVI analyses showed that the L-L and H-H clusters had plain contraction trends to the west and southeast, with decreases of 5.86 % and 14.98 %, respectively; while the Non-significant expressed a clear expansion trend, increasing by 6.90 %; and H-H and Non-significant clusters increased of 14.07 % and 6.90 %, respectively, while L-H cluster showed a significant shrinkage trend of 18.38 %.

3.2.2. Response of NDVI to climate factors

The correlation between NDVI and climate factors was evaluated based on Pearson's correlation coefficient (Pearson's *r*) with the NDVI, temperature and precipitation data of the YSMs in the same period during 1984–2022. The study found that the correlation coefficients of NDVI with temperature and precipitation in the YSMs ranged from $-0.55 \sim 0.76$ and $-0.24 \sim 0.75$ ($P < 0.05$), with the average Pearson's *r* of -0.05 and 0.24, respectively. The area where NDVI was negatively correlated with temperature accounts for 55.79 %, while 30.53 % was positively correlated, and 13.55 % showed a significant negative correlation. Among them, the negative correlation was concentrated in the desert steppe area on the northern slope of the YSMs, the positive correlation was banded in the southern edge of the YSMs, and 13.68 % was distributed in the western part of the region, which was covered no correlation. However, NDVI was highly correlated with precipitation, with a positive correlation of up to 80.05 % of this area, a negative correlation of only 1.27 % and the remaining 13.68 % being no correlation (Fig. 11a–d).

The results of Pearson's correlation showed that the positive correlation between NDVI and precipitation in the YSMs was much larger than that of temperature, *i.e.*, precipitation was the dominant factor affecting vegetation growth in the YSMs. Thus, the inter-annual variation of vegetation index was more sensitive to precipitation.

Further research using Partial correlation analysis, controlling for either temperature or precipitation, was conducted to assess the individual effects of the other variable on NDVI and verify the result of the Pearson's correlation analysis. As shown in Fig. 11e–h, the partial correlation coefficients (Partial's *r*) of NDVI with temperature and precipitation in the YSMs ranged from $-0.54 \sim 0.76$ and $-0.21 \sim 0.75$ ($P < 0.05$), with the average Partial's *r* of -0.04 and 0.25, respectively. The negative and the positive correlation between NDVI and temperature accounted for 52.93 % and 33.40 %, which were concentrated in the northern arid and southern semi-arid regions of the YSMs, respectively. On the contrary, the positive correlation between NDVI and precipitation was as high as 85.05 % of the total area, which was distributed across the entire YSMs, except for the barren in the west.

The joint analysis of Pearson's and Partial's correlation results implied that precipitation was the dominant climatic factor influencing NDVI variations in the YSMs, which showed a high positive correlation with NDVI and largely determined the growth of vegetation. In contrast, there was no obvious regularity in the effect of temperature on NDVI. This finding was consistent with previous researches on vegetation drivers in the YSMs (Fan et al., 2007; Zhang et al., 2024b).

3.2.3. Response of NDVI to human activities

Climate change is an crucial factor affecting vegetation growth, but in recent years, high intensity human activities have led to anthropogenic stress on vegetation growth to some extent (Cao et al., 2018; Shi et al., 2020). In this research, the influence of climatic factors on long-term NDVI changes was excluded based on multiple regression residual analysis to quantify the extent of the influence of non-climatic factors (mainly referring to human activities) on the NDVI of the YSMs (Fig. 12). In addition, the degree of correlation between multiple factors (including temprature, TMP; precipitation, PRE; Population, POP; Nighttime-light, NTL; and grasslands grazing intensity, GGI) was characterized using Pearson correlation coefficient (Fig. 13).

The results of the spatial distribution of the residual results revealed that the NDVI residuals ranged from $-0.0013 \sim 0.0016$, with an average value of -0.0001, indicating that human impacts were very weak (Fig. 12a). Fig. 12b showed the T-S & M-K trend analysis for the residual series from 1984 to 2022, and statistically, a decreasing trend accounted for 59.41 %, while those with an increasing trend constituted 26.91 %. Notably, the largest proportions are in non-significant decreases and increases, at 48.47 % and 23.73 %, respectively. Besides, considering

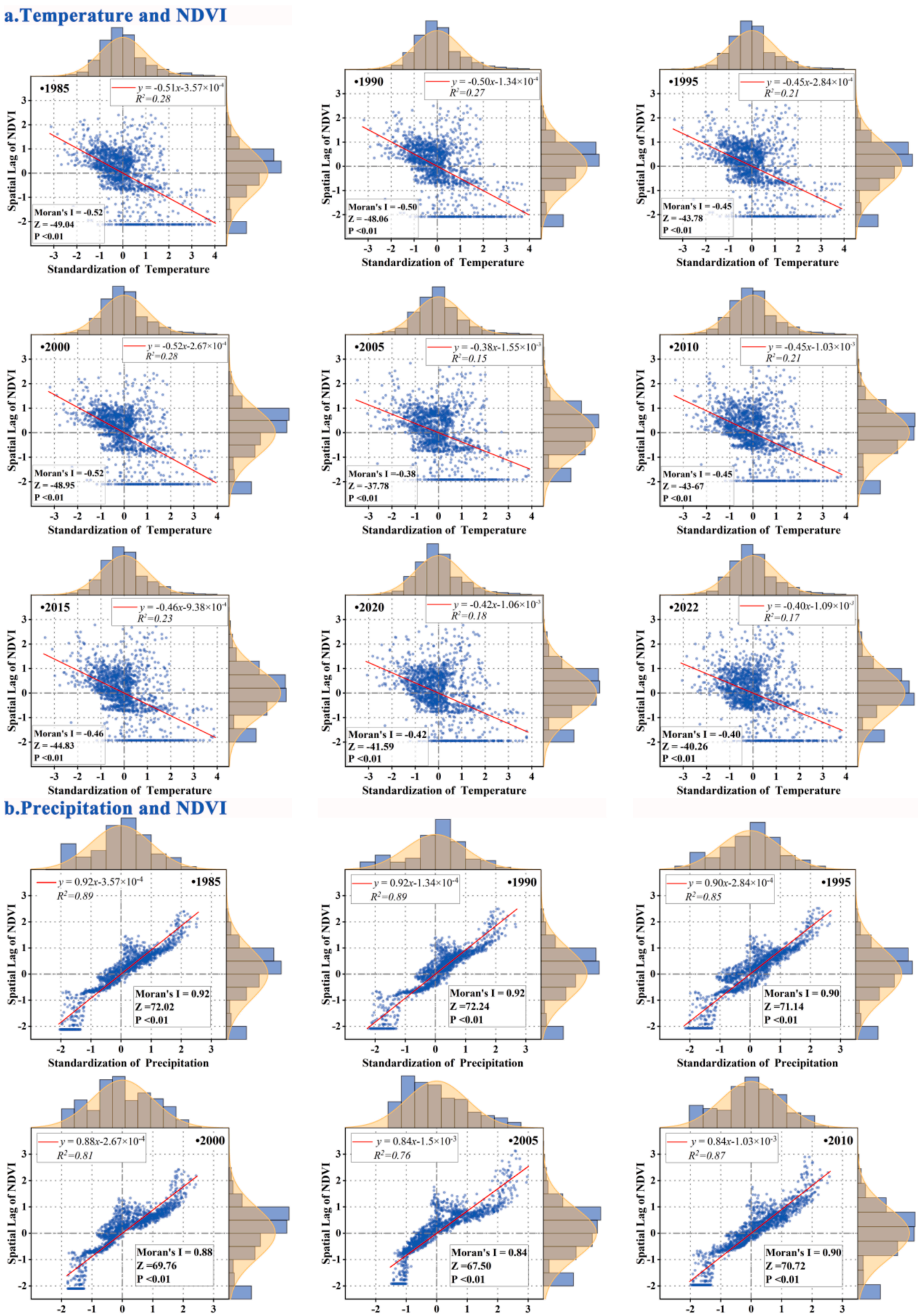

Fig. 8. Scatter plot of Bivariate Global Spatial Autocorrelation (1984–2022). **a.** Temperature and NDVI; **b.** Precipitation and NDVI.

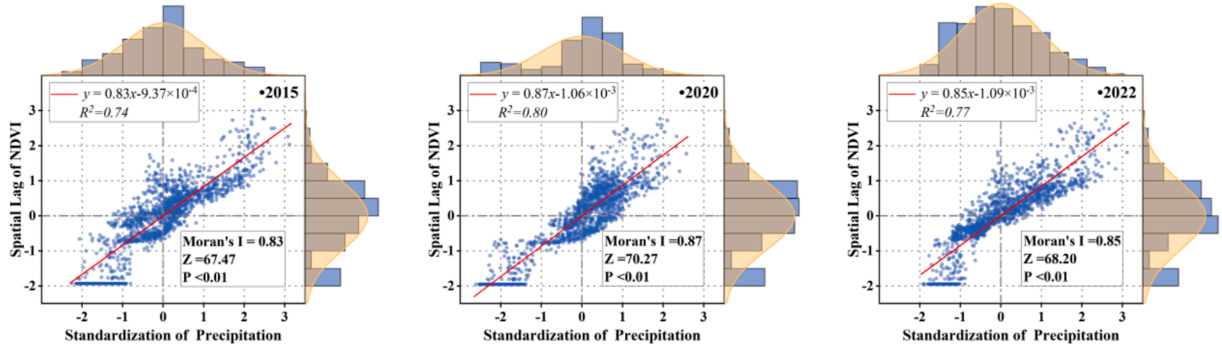

Fig. 8. (*continued*).

that the residual results were only a generalized analysis of human activities, direct quantitative indicators that can characterize human activities (*e.g.,* POP, NTL, and GGI) were included in this paper to refine the analysis of anthropogenic factors(Fig. 13). Due to data limitations, with the grazing intensity dataset only covering the YSMs region in Inner Mongolia, the temporal and spatial scope of the multiple influencing factor correlation analysis is limited to 84 % of the study area from 2000 to 2020. The results showed that, in terms of the influence of vegetation growth, only TMP played an inhibitory role on NDVI, and the correlation was always maintained at around −0.50 ($P < 0.01$). Conversely, PRE and other factors played a promotive role, with the degree of influence ranked as follows: PRE (0.85) > GGI (0.31) > NTL (0.13) > POP (0.10), all of the above were $Pearson'r_{mean}$ with $P < 0.01$.

In summary, it can be concluded that: (1) For climatic factors, PRE was the dominant factor in promoting vegetation growth; (2) For anthropogenic factors, GGI had a greater influence compared to POP and NTL, because the north of YSMs was mostly a nomadic civilization and had fewer built-up areas as well as sparser human populations. (3) Compared to climatic factors, the influence of anthropogenic factors in the region was much smaller than that of climate, but overall, they had a promotive effect. (4) Regarding the fluctuation of influence intensity, the degree of influence of each factor and NDVI has remained relatively stable since 2000, with no significant fluctuations observed.

4. Discussion

4.1. Reliability verification of typical sites

The study found that the NDVI time-series data for the western part of the YSMs, specifically in the Langshan mountain (belonging to the Jilantai Desert), consistently showed 0 values. This led to a lack of significant temporal variation analysis in this area. To ensure the reliability of the analysis results, the research validated the PKU GIMMS NDVI (February 2000–December 2022) based on the resampled MODIS MOD13C2 data.

On the intra-annual scale, the correlation analysis between PKU GIMMS NDVI and MODIS NDVI from 2000 to 2022 showed that the correlation coefficients of $r = 0.98$ ($P < 0.05$), which indicated that the trends of PKU GIMMS NDVI and MODIS NDVI were highly consistent; from Fig. 14a, the linear fitting slope was $Slope = 1.017$, parallel to $y = x$ and located on one side of the MODIS axis, suggesting that there is a systematic underestimation of PKU GIMMS NDVI data compared with MODIS NDVI.

On the inter-annual scale, the average annual values were taken to compare the inter-annual trends of PKU GIMMS NDVI and MODIS NDVI. The results showed that the pattern of change of both datasets was consistent, with an increasing trend at the rates of 7.95×10^{-4}/a ($R^2 = 0.164$) and 1.59×10^{-3}/a ($R^2 = 0.493$), respectively, and the correlation coefficient was $r = 0.75$ ($P < 0.05$). Due to the difference in spatial resolution, the annual PKU GIMMS NDVI values were lower than the annual MODIS NDVI values by 1 ~ 3 % during the same period, as shown in Fig. 14b.

Overall, the trends in PKU GIMMS NDVI and MODIS NDVI were largely consistent, with high correlation on both intra-annual ($r = 0.98$, $P < 0.05$) and inter-annual ($r = 0.75$, $P < 0.05$) scales. The annual values of PKU GIMMS NDVI were underestimated by 1 ~ 3 % compared to MODIS NDVI, the difference was minor. This indicated that it is feasible to use PKU GIMMS NDVI as the foundational data to characterize the long-term vegetation evolution in the region.

4.2. Comparison of peer researches for the YSMs

China is one of the countries with the extensive distribution of ecological vulnerable area (EVA) and the greatest variety of fragile ecosystems in the world (Hu et al., 2021; Sun et al., 2019). It accounts for more than half of the national territory (Deng et al., 2016). Under the dual impacts of global climate change and human activities, the conflicts between human-land relations in the EVA have become increasingly prominent, resulting in ongoing environmental degradation. This has seriously constrained the local sustainable development of the economy and ecology (Huang et al., 2019; Ma et al., 2023; Yang et al., 2021). The YSMs, as a typical area of the EVA in China, is marked by noticeable grassland degradation, fragmentation of migratory birds' habitats and breeding sites (Liu et al., 2015), and serious loss of biodiversity. Studying the changing law of the ecological pattern of this region can provide valuable insights for the ecological governance of other EVA in China.

Currently, there have been a substantial body of research focused on the regional ecological environment of the YSMs. However, most previous researches were still based on the last generation of datasets, such as GIMMS NDVI (Pinzon and Tucker, 2014), MODIS NDVI (Seguini et al., 2024) or SPOT NDVI (Bai et al., 2019), which generally had the problem of shorter time-series. Based on the new generation of long time-series NDVI products released by Peking University (PKU GIMMS NDVI), this paper provided a comprehensive analysis of change patterns and spatial variability of vegetation growth in the YSMs over nearly 40 years. The findings indicated that the overall ecological level of the region is low. After a long period of comprehensive management, the overall deterioration of the eco-environment has been curbed, nevertheless, fully reversing the ecological fragility and sensitivity remains a distant goal; moreover, the previous researchers have paid more attention to the NF-YSMS in terms of spatial domains (Chaoligeer et al., 2024; Wang et al., 2024; Zhang et al., 2024a,b). Taking into account the geomorphological differences between the north and south sides of the YSMs, this paper investigated the entire YSMs, and the results found that significant differences in vegetation growth between the north and south slopes. In view of this, the north and south slopes were divided according to the ridge line extracted from the highest point of the mountain (Fig. 1) to further compare and analyze the vegetation growth on the north and south slopes of the YSMs. As shown in Fig. 15, both past

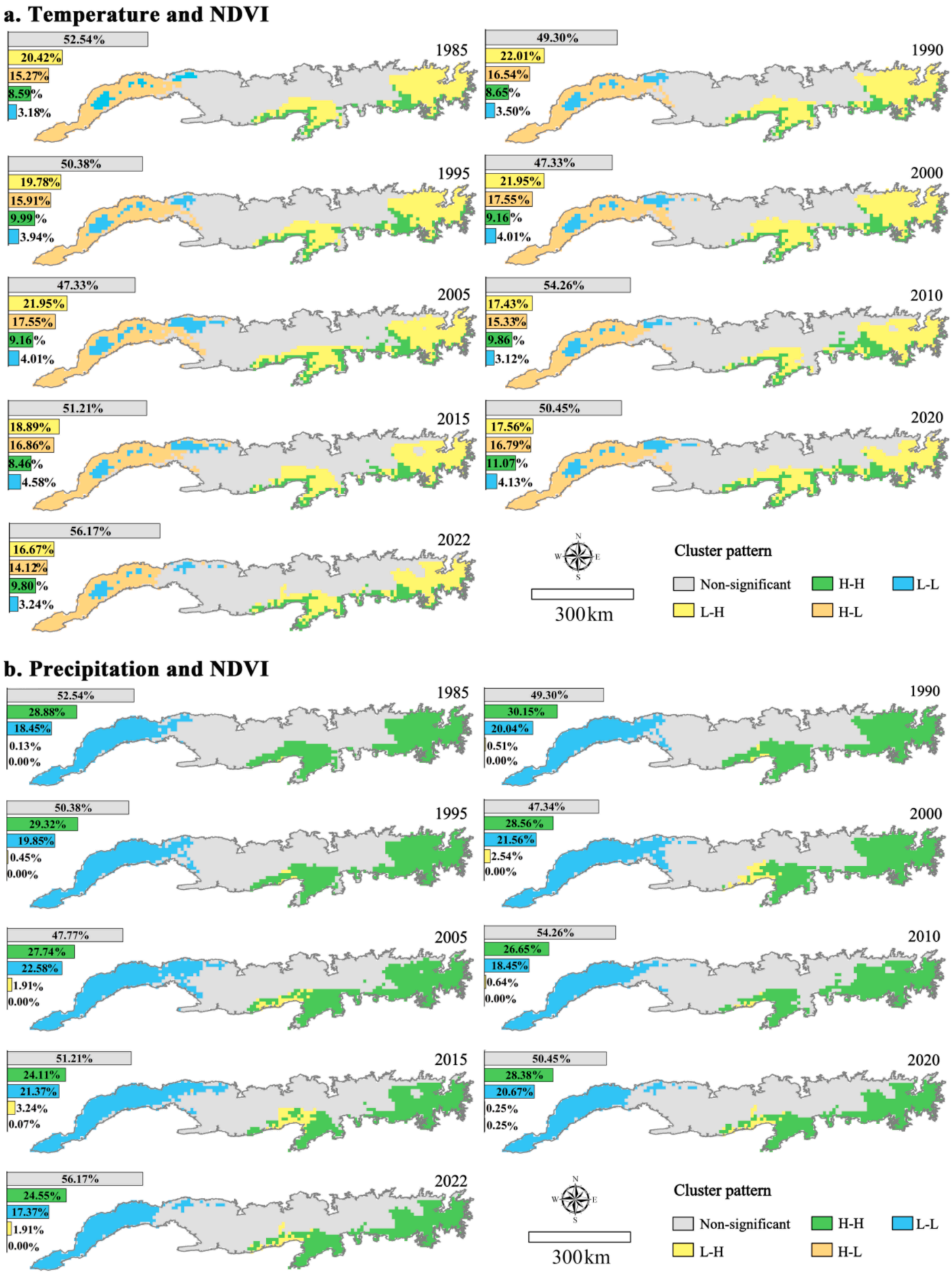

Fig. 9. Spatial clustering patterns between climate factors and NDVI (1984–2022). **a.** Temperature and NDVI; **b.** Precipitation and NDVI.

evolutionary features (Fig. 15a) and future trend projections (Fig. 15b) showed strong spatial heterogeneity on the north and south sides of the ridge line. Separate statistics for both sides show that the north slope was mainly characterized by a decreasing trend (66.07 %) during the last 39 years, while most of the south slope shows an increasing trend, accounting for 62.77 %. At the same time, the Hurst index was used to predict the future trend of the north and south sides respectively, and the results showed that 52.01 % of the area of the north slope will show improvement (*i.e.,* 45.36 % of anti-sustainable decrease & 6.65 % sustainable increase), while more than half of the area of the south slope (53.27 %) would show degradation (*i.e.,* 39.61 % of anti-sustainable increase & 13.66 % sustainable decrease). This result suggested that although the vegetation on the south slope has generally outperformed the north slope in the past, it was highly likely to face extensive

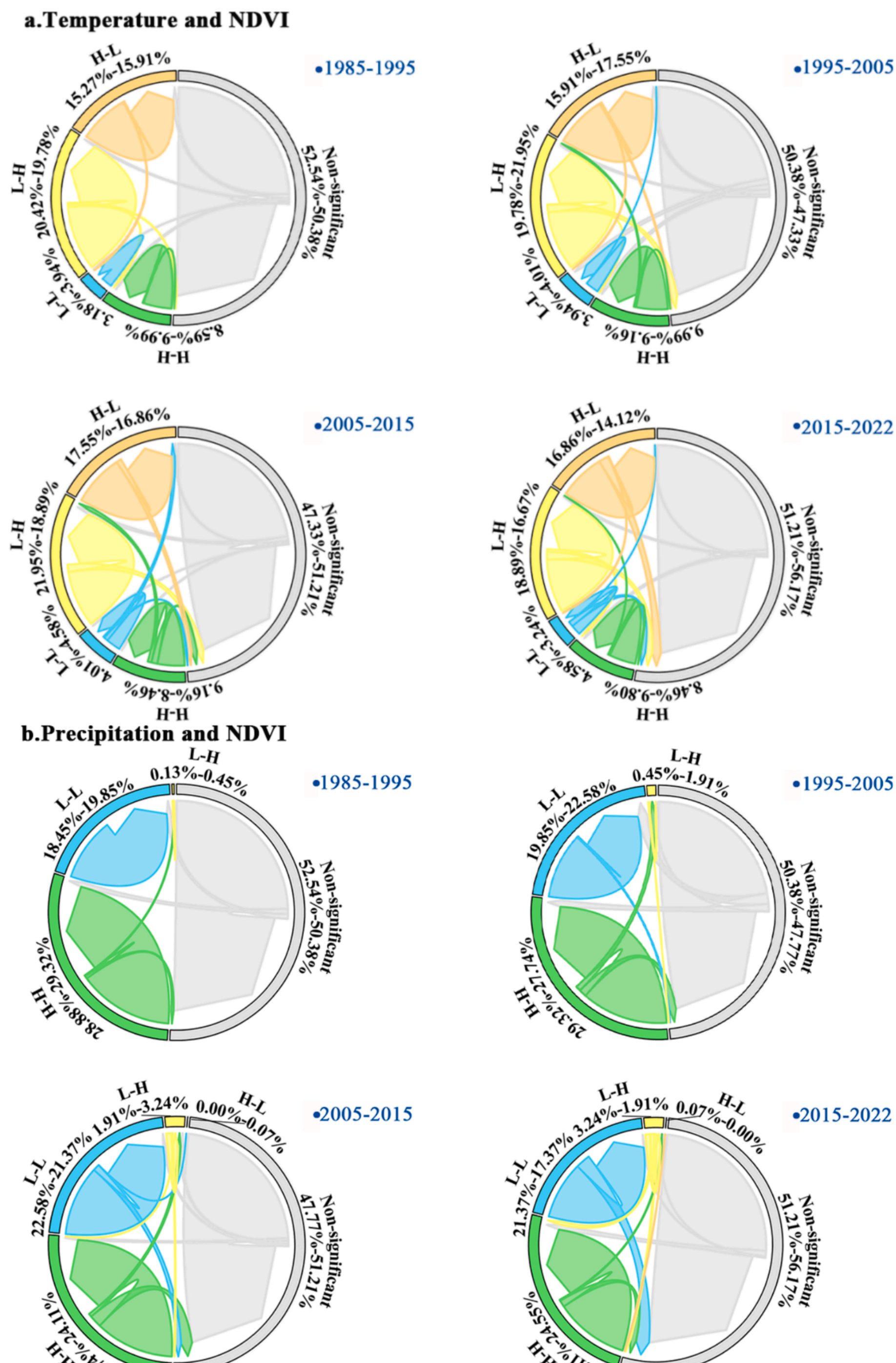

Fig. 10. Type transitions of spatial clustering patterns between climate factors and NDVI (1984–2022). **a.** Temperature and NDVI; **b.** Precipitation and NDVI.

vegetation degradation in the future. The above differences may be related to the differences in vegetation distribution, with temperate deserts and desert steppe dominating in the northern part of the YSMs, and temperate grasslands and forests in the southern part (You et al., 2021). In addition, the Hetao Plain located south of the YSMs was relatively rich in water resources thanks to the irrigation from the Yellow River (Wang et al., 2020), which played a role in nourishing the local ecosystem, and the southern slopes of the YSMs have consequently maintained a better vegetation growth over the past 39 years. However, in recent decades, strong human activities such as land reclamation (Wei et al., 2021), rapid urbanization, and construction of water conservancy projects (Omer et al., 2020; Xie et al., 2019) in the area have also caused some disturbances to the vegetation growth, and the stability of the ecosystem has been damaged. In the future, we should take into full consideration the differences between the two sides of YSMs, and tailor the eco-environment management programs for the local conditions, so as to effectively curb the decline of the vegetation on the southern slope, and consolidate the upward trend of the ecology on the northern slope.

4.3. Influence of climate and anthropogenic factors on vegetation change

The 400 mm isohyet is a very geographically significant wet-dry demarcation line, which divides China into semi-humid and semi-arid climate zones (Qiao and Xia, 2024). After extracting the 400 mm

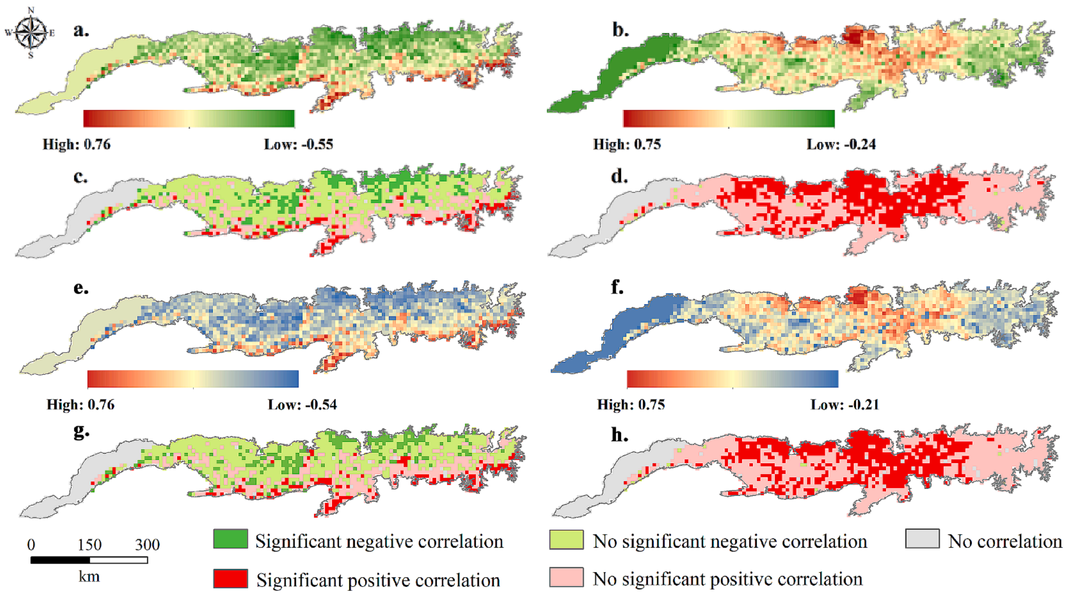

Fig. 11. The correlation analysis and significance tests of average annual NDVI with average annual temperature and total annual precipitation in the Yinshan mountains, 1984–2022.

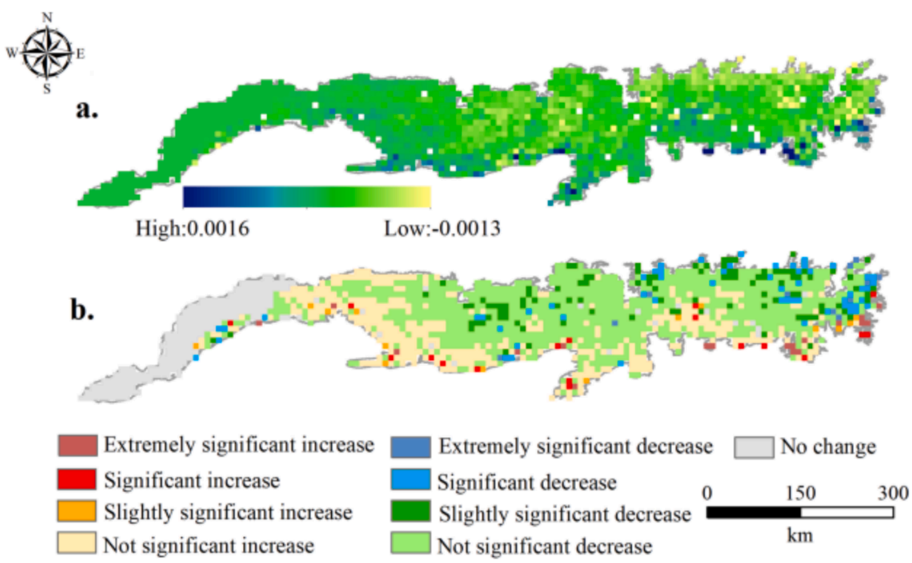

Fig. 12. Residual analysis of NDVI in the Yinshan Mountains, 1984–2022. **a.** Spatial distribution of residual value; **b.** Characteristics of spatial distribution T-S & M-K.

isohyet in this paper (Fig. 1), we found that 91.79 % of YSMs was located north of the 400 mm isohyet (less than 400 mm), indicating that the available water resources in this area were limited, which will directly affect the vegetation growth and the overall function of the ecosystem. Studies indicated that the spatial distribution of precipitation and vegetation growth basically coincides with each other, they had a positive pattern of spatial association (*Moran's* $I = 0.88$, $P < 0.01$). This study showed that the spatial distribution of vegetation growth and precipitation in this region basically coincided with each other, and there was a positive spatial correlation pattern between the two (*Moran's* $I = 0.88$, $P < 0.01$). This was mainly due to the unique mountain trend of the YSMs, *i.e.,* the higher altitude of Daqingshan and Huitengliang mountains were perpendicular to the direction of the Pacific moisture transport routes, thus facilitating the uplift and blockage of moisture (Gong et al., 2023). At the same time, this affirmed that precipitation was the major climatic factor affecting vegetation growth in the study area. This finding aligns with the results of previous studies on the climate response of vegetation in the YSMs (Fan, 2010; Xia et al., 2007a,b). Precipitation, as the first dominant factor in the variation of wet-dry conditions in most parts of China, had a significant effect on the physiological and ecological processes of vegetation (Li et al., 2024; Yuan et al., 2017). It has been shown that the 400 mm isohyet has a tendency to move westward and uplift northward (Gao et al., 2020). As a sensitive zone such as arid-semi-arid and desert-steppe transition zone, the YSMs will be more sensitive to the fluctuation and change of the 400 mm isohyet (Xu et al., 2022; Zhang et al., 2016). This suggests that as rainfall increases in northern China, the vegetation cover in previously desert or semi-desert areas may improve in this region. This undoubtedly presents an excellent opportunity for combating land desertification and enhancing the ecological conditions of the area.

Climatic factors and human activities are considered to be the two most direct drivers of vegetation change (Fang et al., 2021; Wei et al.,

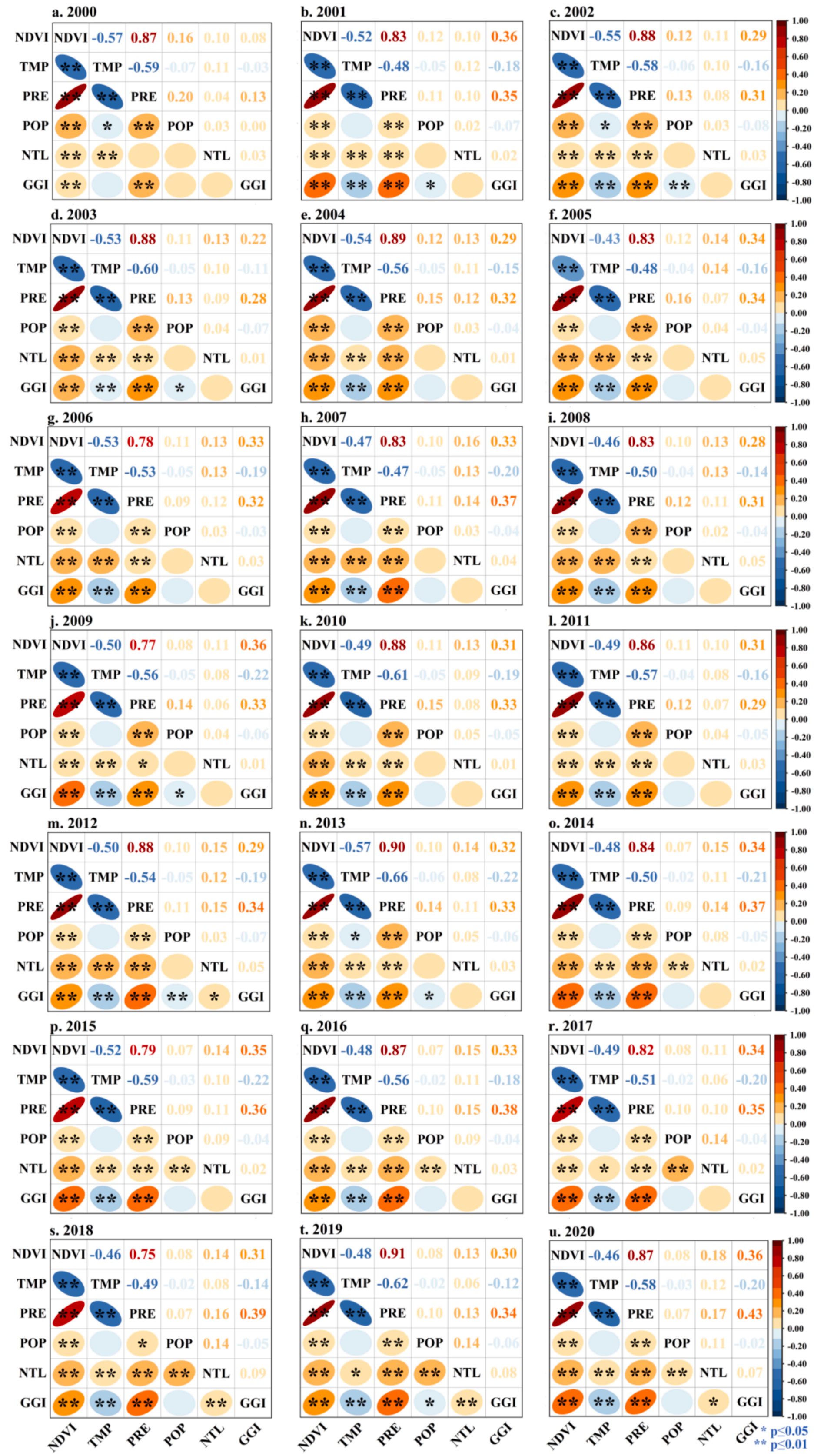

Fig. 13. Correlation analysis of multiple impact factors in the Yinshan Mountains, 2000–2022.

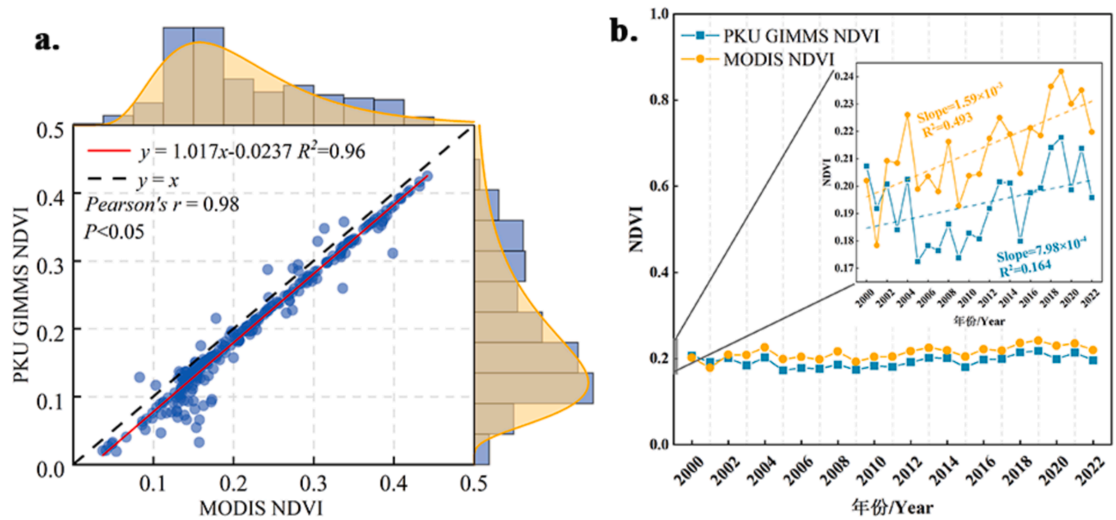

Fig. 14. Consistency test and comparison of trend analysis of PKU GIMMS NDVI and MODIS NDVI (2000–2022).

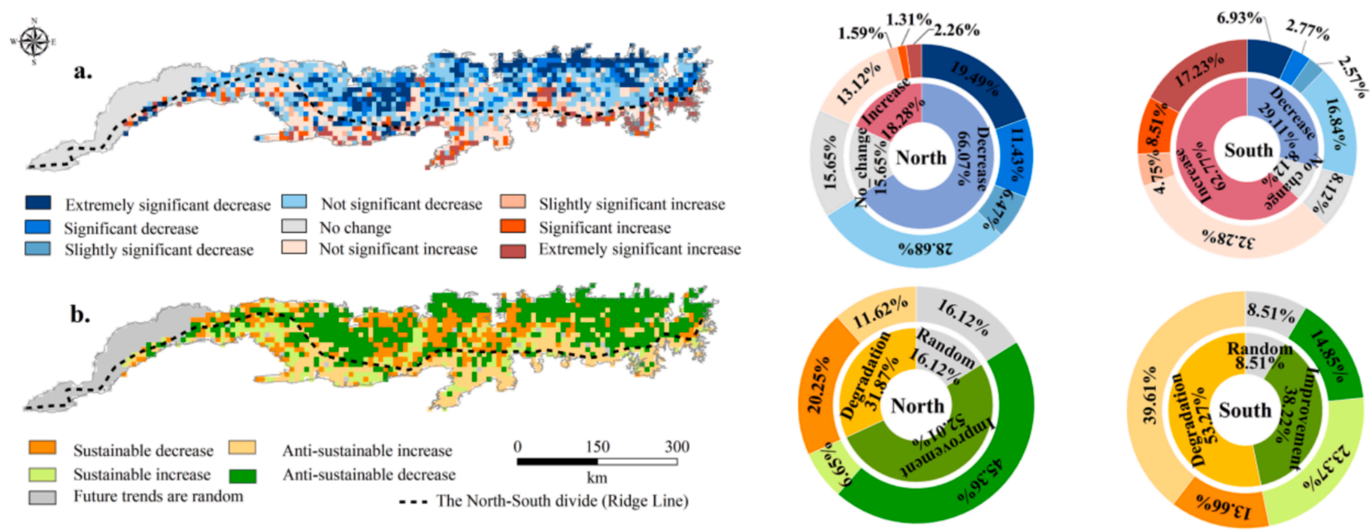

Fig. 15. Comparison of vegetation growth on the north and south slopes of study area.

2022), and assessing the impacts of climate change and human activities on vegetation growth is critical for maintaining ecosystem stability (Chen et al., 2021; Gao et al., 2021). YSMs was located at the southern edge of the Inner Mongolia Plateauand and was situated in a zone where multiple ecological elements interact. The entire mountain was a dry steppe and desert grassland landscape with a long-term low ecological status, therefore, it was more susceptible to climatic and anthropogenic perturbations (Wang et al., 2024; Yang et al., 2023a,b). Inner Mongolia, as one of the four major pastoral areas in China, the increase in grazing has brought great pressure on the steppe ecosystems, and so far, nearly half of Inner Mongolia was facing the problem of grassland degradation (Guo et al., 2021). Many studies have proved that overgrazing also triggers a series of serious consequences, such as the reduction of biodiversity (Lin et al., 2022), the decline of vegetation cover (Chen et al., 2020) and land sands (Dai et al., 2022). Thankfully, under specific conditions, active and reasonable human activities can effectively restore and improve the vitality and vigor of vegetation (Kong et al., 2017; Lu et al., 2024). Moreover, studies have confirmed this conclusion: Mu et al., (2013) pointed out that during the period 2001–2009, the contribution of human activities to grassland greening in Inner Mongolia was as high as 80.23 %; similarly, in the desert grassland areas of Xinjiang, human activities played a dominant role in the restoration of vegetation (Zhang et al., 2018). However, studies on the driving factors of vegetation in the YSMs have primarily focused on climatic factors, especially precipitation. As one of the main natural factors affecting vegetation, vegetation showed a strong response to changes in precipitation (Xia et al., 2007a; Zhang et al., 2024b; Zhao et al., 2017b). Considering the non-negligible role of human activities in the process of vegetation greening, this study quantified the impact of human activities on vegetation growth based on residual analysis. Residual analysis was one of the most widely used methods for accurately assessing the impact of climate-anthropogenic factors on vegetation. It was based on the assumption that the effects of human activities and climatic factors on vegetation growth were independent of each other, so that the effects of human activities can be detected by eliminating climate change (Li et al., 2018). However, precisely because of this, the results of residual analysis only provide a generalized interpretation of human activities. Based on residual analysis, the population distribution, nighttime-light and grazing intensity were used to test the impact of specific human daily production and life on vegetation in this study. Both analyses suggested that the correlation between human activities and NDVI, although low, showed an overall facilitating effect after 2000. Furthermore, the YSMs, as a zone where nomadic civilization and farming civilization are intertwined, the extensive grasslands to the north provide ample grazing space for nomadic peoples. However, the research showed that grazing intensity didn't exhibit a negative impact; rather, it contributed a slight positive effect on vegetation growth. In conclusion, because of the YSMs as an EVA, the resonance and coupling effects of global climate change and human activities still have the potential to exacerbate the sensitivity of the ecological environment of the region to external disturbances, and therefore the impacts of human activities in the region should not be ignored despite their weak impacts.

4.4. Limitations and recommendations

First, in this study, the PKU GIMMS NDVI data were selected to examine the spatiotemporal evolution characteristics of vegetation in the YSMs. It has a large time span but a low spatial resolution (8 km). Therefore, the results of the study only provided a macroscopic explanation of the evolution pattern of the vegetation and the response mechanism owing to the limitations of the data itself. Meanwhile, it was not possible to conduct more detailed analyses the different land-use types, vegetation types, and other specific areas within the region. Future research propose to utilize higher spatial resolution multi-source geographic information data and combine population, economy, and grazing data to carry out more detailed research and examination of the area. Secondly, this paper adopted the "Nearest Neighbor Interpolation Method" to interpolate the meteorological data and land use data to fit the spatial resolution of NDVI products for subsequent operations. However, due to the large difference in spatial resolution between NDVI products and other data (8 km vs. 1 km/30 m), small-scale features or phenomena may be lost in the resampling process, which was mainly manifested in this paper by the greatly reduced accuracy of the extraction of impervious surfaces in the case of 30 m to 8 km down-sampling; in addition, the 8 km scale of the resampled data resulted in the averaging of features, which can not capture the local features that can be characterized by meteorological or anthropogenic data at the 1 km scale precisely, it clearly increased the limitations and uncertainties of driven analyses.

5. Conclusions

The study investigated the spatiotemporal evolution characteristics and spatiotemporal response of NDVI to various driving factors in the YSMs over 39 years (1984–2022) based on PKU GIMMS NDVI, meteorological (precipitation and temperature), land cover, and DEM data. The main findings are as follows:

(1) As an ecological barrier in the ASRs of northern China, the vegetation in the YSMs was characterized by strong spatial heterogeneity of PKU GIMMS NDVI. The $NDVI_{max}$ ranged from 0.46 to 0.53, generally decreasing from southeast to northwest, indicating a relatively fragile ecological environment. Over the 39-year period, the vegetation growth has gradually improved and shown phased development, especially in the past 20 years, showing a clear trend of improvement ($Slope = 1.56 \times 10^{-3}/a$). This fully demonstrates the effectiveness of ecological projects such as the Three North Shelter Forest System, Grain to Green and Natural Forest Protection.

(2) As a climate barrier in the ASRs of northern China, a systematic analysis of 39 years of climate change in the region and the driving mechanisms, revealed that the temperature and precipitation in the region have been rising at a fluctuating rate of an average of 0.45 °C/10a and 5.58 mm/10a, respectively. This serves as a warning signal of a transition towards "warmer and wetter" conditions, representing a strong regional response to global climate change.

(3) The response of NDVI to climate change and human activities in the YSMs was a dynamically complex and interactive process. The correlation analysis revealed that the sensitivity of NDVI to precipitation was higher than that of temperature in this area, and the spatial pattern of vegetation growth and precipitation exhibited high clustering characteristics ($Moran's\ I = 0.88,\ P < 0.01$); among them, H-H and L-L clusters were dominant feature, and there existed an obvious characteristic of zonal differentiation: "more in the south, less in the north; higher in the east, and lower in the west". Additionally, anthropogenic activities consistently showed a small correlation with NDVI, but weak impacts may elicit a sensitive response from the vegetation, suggesting that tthe overall anti-disturbance ability and stability were weak.

(4) The overall sustainability of vegetation NDVI was in-apparent in the YSMs. Over the 39-year period, only 28.05 % of the area exhibited a Hurst index greater than 0.5, indicating a trend towards inverse vegetation growth in the future. Additionally, degradation was primarily concentrated in the southern regions of the YSMs, such as Wulashan mountain, Manhanshan mountain, and Huitengliang mountain, suggesting that the ongoing improvement in the southern region may reverse. It is necessary to prevent the southward movement of the desert-steppe transition zone in a timely and dynamic interventions manner.

In conclusion, the evolutionary characteristics of vegetation in the YSMs were characterized by stage, spatiotemporal heterogeneity and zonal differentiation. Its evolutionary mechanism was complex and multifaceted, which is the result of intersecting response to climate change and human activities. Future research may point to non-linear residual analysis models on smaller-scale so as to propose more targeted vegetation management schemes.

CRediT authorship contribution statement

Menglin Yan: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Jingyang Lu:** Writing – review & editing, Visualization, Formal analysis, Data curation. **Yingying Ma:** Writing – review & editing, Validation, Formal analysis. **Chao Ma:** Writing – review & editing, Visualization, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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